

Create two new EC2 instance with Ubuntu OS. ansible-server and target-server. Open ansible-server using MobaXterm.

1. **sudo apt update** to update package manager.

```
ubuntu@ip-172-31-24-44:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
[137 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
[136 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages
[16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en
[6329 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components
[4556 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages
[284 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata
[313 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages
[291 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en
[127 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components
[50.0 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata
[8276 B]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages
[340 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en
[89.4 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components
[53.1 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata
[5756 B]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe
```

2. **sudo apt install ansible** to install ansible. There are many ways to install ansible. But it is the best way to install ansible through package manager.

```
ubuntu@ip-172-31-24-44:~$ sudo apt install ansible
Installing:
  ansible

Installing dependencies:
  ansible-core          python3-kerberos      python3-selinux
  python3-argcomplete   python3-libcloud      python3-winrm
  python3-decorator      python3-pyspnego      python3-xmltodict
  python3-dnspython      python3-requests-ntlm
  python3-gssapi          python3-resolvelib

Suggested packages:
  cowsay  python3-aioquic  python3-httpcore  python3-trio
  sshpass  python3-h2  python3-httpx

Summary:
  Upgrading: 0, Installing: 14, Removing: 0, Not Upgrading: 66
  Download size: 22.4 MB
  Space needed: 245 MB / 4679 MB available

Continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 python3-resolvelib all 1.2.1-2 [26.1 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 python3-dnspython all 2.8.0-1ubuntu1 [178 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 ansible-core all 2.20.1-1 [1379 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 ansible all 13.1.0+dfsg-1ubuntu1 [19.1 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 python3-argcomplete all 3.6.3-1 [36.1 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 python3-decorator all 5.2.1-2 [28.1 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 python3-gssapi amd64 1.10.1-1build1 [503 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 python3-kerberos amd64 1.1.14-3.1build13 [20.9 kB]
```

3. **ansible --version** to verify whether ansible is installed correctly.

```
ubuntu@ip-172-31-24-44:~$ ansible --version
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.14.4 (main, Apr  8 2026, 04:02:31) [GCC 15.2.0] (/usr/bin/python3)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
ubuntu@ip-172-31-24-44:~$
```

SETUP PASSWORLESS AUTHENTICATION

1. **ssh-keygen** in ansible-server.

```
ubuntu@ip-172-31-24-44:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519): /home/ubuntu/.ssh/id_ed25519
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:N6P8LQ0G15EmDULg4qL5es+j2R+kWxDN0HRJR1cno60 ubuntu@ip-172-31-24-44
The key's randomart image is:
+--[ED25519 256]--+
|    o++oo ..+ .  |
|    o =o... o +  |
|    . = o. o...  |
|    . . o . =.   |
|    . . . .S+E.  |
|.o  +..+o.o     |
|o   . 000.      |
|..+.o . .o.     |
|.o+o=o. .o.     |
+----[SHA256]-----+
ubuntu@ip-172-31-24-44:~$
```

This will generate a key(public & private). It created set of files.

2. **ls /home/ubuntu/.ssh/** to see what are the files created.

```
ubuntu@ip-172-31-24-44:~$ ls /home/ubuntu/.ssh/
authorized_keys  id_ed25519  id_ed25519.pub  known_hosts
ubuntu@ip-172-31-24-44:~$
```

Private key is in the id_ed25519 file. We can use it to log into the machine. **Never share** it.

Public key is in the id_ed25519.pub. If someone wants to communicate with each other always go with the public key.

3. **cat /home/ubuntu/.ssh/ id_ed25519.pub**

```
ubuntu@ip-172-31-24-44:~$ cat /home/ubuntu/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIE3GxVEvNm2oKHBSDJvLc122DyTXKY33hzfSynLgweLA ubuntu@ip-172-31-24-44
```

Copy this public key.

Then take a new tab on MobaXterm. Log into the target-server. Do ssh-keygen here as well.

4. ssh-keygen

```
ubuntu@ip-172-31-19-182:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519): /home/ubuntu/.ssh/id_ed25519
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
```

It will also create a set of files like before.

5. **ls ~/.ssh/** to see file list.

```
ubuntu@ip-172-31-19-182:~$ ls ~/.ssh/
authorized_keys  id_ed25519  id_ed25519.pub
ubuntu@ip-172-31-19-182:~$
```

6. **vim ~/.ssh.authorized_keys** to open authorized_keys

Paste the earlier copied content.(public key)

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIE3GxVEvNm2oKHBSDJvLc122DyTXKY33hzfSynLgweLA ubuntu@ip-172-31-24-44
~
~
~
~
~
```

7. Go back to ansible-server. **ssh -private IP of target-server-**

Now we can access target-server from ansible-server with password less authentication.

```
ubuntu@ip-172-31-24-44:~$ ssh 172.31.19.182
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Thu Jul 16 17:24:26 UTC 2026

System load:  0.0           Temperature:    -273.1 C
Usage of /:   31.5% of 6.61GB Processes:      118
Memory usage: 28%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.19.182

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu Jul 16 16:41:32 2026 from 43.250.241.41
ubuntu@ip-172-31-19-182:~$
```